

AI Strategy Proposal

AI Quality Incident Assistant for Hertzmann & Söhne

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Executive Summary

Hertzmann & Söhne operates in the automotive manufacturing sector, where quality documentation plays an essential role in maintaining compliance with standards such as ISO 9001 and IATF 16949. Quality engineers regularly prepare Non-Conformance Reports (NCRs) to document production and supplier-related quality incidents. These reports require engineers to review inspection notes, identify the relevant information, organize it into a standardized format, and ensure that all required fields are completed before the report can be reviewed and approved.

Although the process is essential, much of the work is repetitive and administrative. Engineers spend valuable time manually transferring information from handwritten or digital inspection notes into standardized documentation rather than focusing on investigation and problem solving. Differences in reporting style may also lead to inconsistencies and missing information that require additional review.

This proposal recommends implementing an **AI Quality Incident Assistant** that supports quality engineers during the documentation process. The assistant uses a Large Language Model (LLM) to transform unstructured quality incident notes into a structured first draft of a Non-Conformance Report. The system extracts factual information, identifies missing fields, assigns a preliminary incident category, and prepares a standardized report for review.

The proposed solution is intended only as a documentation support tool. It does not determine product compliance, identify root causes, approve corrective actions, or replace engineering judgement. Every generated report must be reviewed and approved by a qualified quality engineer before becoming part of the quality management system.

A controlled pilot project using historical anonymized quality incidents is recommended before any broader implementation.

1. Business Problem

Quality documentation is an essential part of manufacturing operations. Every production or supplier-related quality issue must be documented accurately to support traceability, regulatory compliance, and continuous improvement.

Currently, quality engineers manually prepare Non-Conformance Reports using information collected from inspection notes and production records. This process requires engineers to:

- review handwritten or digital inspection notes,
- identify relevant technical information,
- organize information into standardized templates,
- verify report completeness,
- prepare documentation for approval.

While these tasks are necessary, they are largely administrative and consume time that could otherwise be dedicated to technical investigation and process improvement.

Manual documentation also introduces variability between reports. Missing information, inconsistent terminology, and formatting differences can increase review time and require additional revisions before approval.

Reducing the administrative effort associated with quality documentation would allow engineers to spend more time on higher-value activities while improving consistency across reports.

2. Proposed AI Use Case

The proposed solution is an **AI Quality Incident Assistant** that supports engineers during preparation of Non-Conformance Reports.

Instead of creating reports manually, engineers provide their incident notes to the AI assistant.

The assistant then:

- extracts relevant facts,
- identifies product identifiers,
- captures measurements and tolerances,
- records containment actions,
- identifies missing information,

- assigns a preliminary incident category,
- produces a standardized draft report.

The draft is then reviewed, corrected if necessary, and approved by a qualified quality engineer.

The workflow is illustrated below.

Quality Engineer

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Raw Quality Incident Notes

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AI Quality Incident Assistant

- Extract facts
- Classify incident
- Detect missing information
- Draft NCR

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Quality Engineer Review

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Approved Non-Conformance Report

The AI assistant functions solely as a documentation aid and does not replace existing quality management procedures.

3. Why This Use Case?

This use case is well suited for generative AI because the task primarily involves understanding natural language and organizing existing information rather than generating new technical knowledge.

The source information already exists within engineer notes. The AI simply converts this information into a standardized format.

Unlike applications that require autonomous decision-making, this solution keeps all critical engineering decisions under human control while reducing repetitive documentation work.

The use case is therefore considered a low-risk entry point for introducing generative AI into quality management.

4. Recommended AI Tool

After evaluating several commercially available AI assistants, Claude is recommended for the pilot implementation.

Claude was selected because it offers:

- strong performance on long technical documents,
- consistent structured outputs,
- support for XML-style prompt organization,
- reliable instruction following,
- clear formatting suitable for standardized reports.

The pilot can initially be performed using the Claude Pro subscription without software development or system integration.

If the pilot proves successful, future implementation could explore API integration with existing Quality Management Systems.

5. Estimated Business Value

The proposed solution is expected to improve documentation efficiency rather than automate engineering decisions.

Potential benefits include:

- reduced time spent preparing Non-Conformance Reports,
- more consistent report formatting,
- improved completeness of documentation,
- easier identification of missing information,

- increased engineer productivity.

For this proposal, it is assumed that quality engineers currently spend approximately **2–3 hours per week** preparing quality documentation.

If AI reduces drafting time by approximately **40%**, each engineer could save approximately **3–5 hours per month**, allowing additional time for quality investigations and process improvement activities.

These figures represent preliminary estimates intended to demonstrate potential business value. Actual performance should be measured during the pilot project.

6. Pilot Implementation

A limited pilot is recommended before broader deployment.

The pilot should include:

- one quality department,
- one standardized NCR template,
- one selected AI model,
- anonymized historical quality incidents,
- mandatory human review.

The pilot should evaluate:

- report preparation time,
- extraction accuracy,
- consistency of report structure,
- identification of missing information,
- user satisfaction among quality engineers.

Results should be reviewed before considering additional automation or integration with existing quality systems.

7. Governance and Risk Management

Although the proposed solution presents relatively low operational risk, appropriate governance remains essential.

The AI assistant should never:

- approve reports,
- determine regulatory compliance,
- identify final root causes,
- authorize corrective actions,
- replace engineering judgement.

Quality engineers remain fully responsible for reviewing, editing, and approving every generated report.

Company data should only be processed using enterprise-approved AI services that comply with internal information security policies and applicable data protection regulations.

Historical quality incidents used during testing should be anonymized whenever possible.

8. Example Prompt

The following prompt was developed for the proposed AI assistant.

You are a quality-documentation assistant supporting Hertzmann & Söhne, a German precision-parts manufacturer.

<context>

Hertzmann prepares non-conformance reports as part of its ISO 9001 and IATF 16949 quality processes. Engineers currently prepare these reports manually from inspection and incident notes.

Your output is only a first draft. A qualified quality engineer will verify all facts, confirm or correct the incident category, complete missing information, and approve the final report.

</context>

<source_notes>

{{Paste the engineer's raw incident notes here}}

</source_notes>

<instructions>

Read the source notes and perform the following tasks:

1. Extract only facts explicitly stated in the notes.

2. Assign one preliminary incident category from this list:

- Dimensional deviation

- Material issue

- Surface defect

- Process deviation

- Documentation issue

- Packaging issue

- Other

3. Identify information that is missing from the notes.

4. Create a structured first draft of the quality incident report.

Extract the following fields:

- Part or product identifier

- Batch or lot number

- Detection stage

- Incident category

- Non-conformance description

- Required specification or tolerance

- Actual measured value

- Immediate containment action

- Follow-up explicitly mentioned

- Missing information

Do not invent missing facts.

Do not determine the root cause.

Do not assign final severity.

Do not decide whether the product is compliant.

Do not create corrective actions that are not stated in the notes.

Do not change numerical values, units, part numbers, or batch numbers.

If a value is unavailable, write "Not provided."

</instructions>

<output>

Return one plain-text table with exactly two columns:

Field | Extracted Information

Use exactly these rows:

Part or Product Identifier

Batch or Lot Number

Detection Stage

Preliminary Incident Category

Non-Conformance Description

Required Specification or Tolerance

Actual Measured Value

Immediate Containment Action

Follow-up Mentioned in Notes

Missing Information

After the table, add one line only:

Engineer Review Required: Yes

Do not include an introduction.

Do not include recommendations.

Do not include conclusions.

Do not include commentary outside the table and review line.

</output>

9. Prompt Testing

The proposed prompt was evaluated using three fictional quality incidents representing common manufacturing situations.

The three scenarios included:

Test Case 1

Dimensional deviation

Purpose:

Verify preservation of measurements and dimensional tolerances.

Test Case 2

Documentation issue

Purpose:

Verify recognition of missing supplier documentation without physical product defects.

Test Case 3

Surface defect

Purpose:

Verify classification of cosmetic defects while avoiding unsupported technical conclusions.

Across all three scenarios, the assistant successfully:

- preserved technical identifiers,
- maintained numerical measurements,
- identified missing information,
- assigned appropriate preliminary incident categories,
- followed the requested report structure,
- avoided generating unsupported facts.

Complete testing documentation is provided in the accompanying **Tested Prompt and Output** report.

10. Limitations

The proposed assistant should be viewed as a documentation support tool rather than an autonomous quality system.

The assistant cannot:

- validate engineering measurements,

- determine compliance,
- perform root-cause analysis,
- replace engineering expertise,
- approve final reports.

Complex quality investigations will continue to require expert human judgement.

11. Recommendation

The proposed AI Quality Incident Assistant represents a practical and low-risk opportunity for introducing generative AI into Hertzmann & Söhne's quality documentation process.

The solution focuses on reducing repetitive administrative work while maintaining human oversight for all engineering and regulatory decisions.

A controlled pilot using historical anonymized quality incidents is recommended to evaluate improvements in documentation efficiency, report consistency, and user acceptance before considering broader implementation.